

Follow the instructions below:

1. The exam must be solved individually.
2. Submitting materials retrieved from the internet is plagiarism.
3. You may use all notebooks posted on the course web site and your own notebooks.
4. Solutions must be in a Jupyter notebook, with a SageMath kernel.
5. The questions are provided in a Jupyter notebook.
You may use that notebook to formulate your answers to the questions.
6. Answers must be submitted before, or at 1:50pm.
7. Submit to gradescope.
8. Not submitting any answers will by default result in a zero score.
9. During the exam no questions will be answered, so do not ask questions.

Good Luck!

1. Use the first 20 convergents of the continued fraction of $\sqrt{\pi/3}$ to define the first 20 rational approximations of $\sqrt{\pi/3}$.

Compute the relative errors of those 20 rational approximations to verify the accuracy.

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2. Consider $p = x^4 + 2x^2 + 2x + 1$ as a polynomial with coefficients in the finite field $\mathbb{Z}_3 = \{0, 1, 2\}$, where the arithmetic happens modulo 3.

Add sufficiently many formal roots to this finite field so that p can be written as the product of four linear factors. What are the roots in the extended field?

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3. Explain the difference between sequential and simultaneous substitution.

Illustrate the difference with an example from the course.

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4. Consider the function $f(n) = \frac{2}{n} \sum_{k=1}^{n-1} \left(\frac{k}{n}\right)^2 \cos\left(\frac{k}{n}\right)$.

- (a) Code the definition of f into a list comprehension to make a function F which for some positive integer n returns the floating-point approximation of $f(n)$ in double precision.
- (b) Time the execution of F for $n = 10,000$. Explain why F is inefficient.
- (c) Use Cython to define the function C which returns the floating-point value of $F(n)$. Verify that C is correct. For $n = 10,000$, compare the running time of C with F .

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5. Draw the expression tree of $x - \frac{y^2}{2x + y^2 - 3}$.

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6. Let $p = x^7 - 2x^6 - x^5 - 7x^4 - 2x^3 - 5x^2 + 7x + 2$ be an expression in x .

- (a) What is the best form to numerically evaluate p in the fastest way?
- (b) Illustrate your answer with timings of the evaluation of this form at a random number, comparing with the evaluation of p , given as a symbolic expression.

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7. Rewrite $p = x^3yz^3 + x^2yz^3 + 2yz^4 + x^3 - 2y^2z$

into $q = 2z^4y + z^3yx^3 + z^3yx^2 - 2zy^2 + x^3$ *without* typing q .

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